

Problem B, Cell 3 (A General Upper Bound): partial submission

Lower bound $D_B(T_k - 1) \geq k^2 - 2k - 1$ for every $k \geq 3$

Team submission (Scribe B-C3-014), from gated artefact B-C3-002

Statement (verbatim)

Definitions (verbatim, problem statement). A *partition* of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. Think of it as a division of n cards into s piles. The *shift* $B(\lambda)$ is the partition of n obtained as follows: remove one card from every pile, discard the piles that become empty, and add one new pile consisting of the s removed cards. Formally, $B(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ *cyclic* if $B^i(\lambda) = \lambda$ for some $i \geq 1$, and let

$$d_B(\lambda) = \min\{i \geq 0 : B^i(\lambda) \text{ is cyclic}\}, \quad D_B(n) = \max\{d_B(\lambda) : \lambda \text{ is a partition of } n\}.$$

Write $T_k = \frac{k(k+1)}{2}$, $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the *rank* of n .

C3: A General Upper Bound (verbatim). Now the numbers strictly between two consecutive triangular numbers.

- (a) Prove that for every $k \geq 4$ and every non-triangular n with $T_{k-1} < n < T_k$,

$$D_B(n) \leq k^2 - 2k - 1.$$

- (b) Determine $D_B(T_k - 1)$ exactly.
(c) Determine also, for that n , which partitions attain the maximum.

Status

- **Cell status:** PARTIAL.
- **Claim status of the established claim:** PROVED.
- **Established.** (b) LOWER half: for every $k \geq 3$, $\lambda_k^* = (k-1, k-2, k-2, k-3, \dots, 2, 1, 1)$ is a partition of $T_k - 1$ with $d_B(\lambda_k^*) = k^2 - 2k - 1$, hence $D_B(T_k - 1) \geq k^2 - 2k - 1$ (Sections 0–6 below; PROVED).
- **Remaining gap (OPEN).** (a) the upper bound $d_B \leq k^2 - 2k - 1$ for every non-triangular n of rank $k \geq 4$; the upper half of (b) (so $D_B(T_k - 1) = k^2 - 2k - 1$ is *not* established); (c) the maximiser set. All three are OPEN. See the section “Open parts”.

Cited versus own work

- **Cited.** The Cell 1 classification of cyclic partitions (the earlier cell C1 of this problem, gated earlier in this run; it is used as a cited result and is not re-proved here). In the form used below: for n of rank k , $n = T_{k-1} + r$ with $1 \leq r \leq k$, the cyclic partitions of n are exactly

$$\lambda(e) = (k - 1 + e_1, k - 2 + e_2, \dots, 1 + e_{k-1}, e_k), \quad e \in \{0, 1\}^k, \sum e = r$$

(zero entries omitted). It is used only in Lemma R2 (Section 2.5).

- **Own work (team).** Sections 0–6 (Lemmas R1–R6): the rotation lemma, the energy characterisation of cyclicity, the classification and exact dynamics of partitions with energy excess $\Phi = 1$, and the resulting lower bound. Source: accepted artefact `proof.md` (task B-C3-002), sections 0–6, gated VALID by two referees (B-C3-008, B-C3-009); sha256 `e86838d6c354e239e3201105542c8b83e77e6aa312dfc62488c6a18818c86c7b`.

0 Notation

0.1. We identify a partition $\lambda = (\lambda_1, \dots, \lambda_s)$ with its diagram $C(\lambda) = \{(i, j) \in \mathbb{N}^2 : 0 \leq j < \lambda_{i+1}\}$. Rows $i = 0, \dots, s - 1$ and columns j are 0-based. A finite set $S \subset \mathbb{N}^2$ is the diagram of a partition iff it is an *order ideal*: $(i, j) \in S$ implies $(i - 1, j) \in S$ when $i \geq 1$ and $(i, j - 1) \in S$ when $j \geq 1$. Its rows are then initial segments, and their lengths are weakly decreasing.

0.2. The *diagonal index* of (i, j) is $i + j$. Diagonal d consists of the $d + 1$ positions $(i, d - i)$, $0 \leq i \leq d$; we call i the *row* of the position. $D_m = \{(i, j) : i + j \leq m\}$ is the diagram of δ_{m+1} ($D_{-1} = \emptyset$).

0.3. The energy is $E(S) = \sum_{(i,j) \in S} (i + j)$, and $E(\lambda) = E(C(\lambda))$.

0.4. Throughout, $n = T_{k-1} + r$ with $k \geq 2$ and $1 \leq r \leq k$; this k is the rank of n .

1 Rotation lemma (R1)

1.1. Define R on $\mathbb{N}^2$ by $R(i, j) = (i + 1, j - 1)$ if $j \geq 1$, and $R(i, 0) = (0, i)$. It preserves $i + j$. On diagonal d it sends row i to row $i + 1$ for $i < d$ and row d to row 0, i.e. row $i \mapsto (i + 1) \bmod (d + 1)$. So R is a bijection of each diagonal, hence injective on $\mathbb{N}^2$ and energy-preserving: $E(R(S)) = E(S)$.

1.2. Let λ have s parts and $C = C(\lambda)$. Row 0 of $R(C)$ is $\{(0, i) : (i, 0) \in C\} = \{(0, 0), \dots, (0, s - 1)\}$, of length s . For $0 \leq i \leq s - 1$, row $i + 1$ of $R(C)$ is $\{(i + 1, j - 1) : 1 \leq j < \lambda_{i+1}\}$, i.e. columns $0, \dots, \lambda_{i+1} - 2$, of length $\lambda_{i+1} - 1$. So every row of $R(C)$ is an initial segment, and the row lengths in order are $\ell = (s, \lambda_1 - 1, \dots, \lambda_s - 1)$.

Lemma R1 (1.3). (i) If $\lambda_1 \leq s + 1$, then ℓ is weakly decreasing, $R(C)$ is an order ideal, and $R(C) = C(B(\lambda))$. So $E(B(\lambda)) = E(\lambda)$. (ii) If $\lambda_1 \geq s + 2$, then $R(C)$ is not an order ideal and $E(B(\lambda)) \leq E(\lambda) - 1$.

Proof. (i) $s \geq \lambda_1 - 1 \geq \lambda_2 - 1 \geq \dots \geq \lambda_s - 1 \geq 0$. So the rows (initial segments) have weakly decreasing lengths, which makes $R(C)$ an order ideal. The zero-length rows come last. By definition $B(\lambda)$ is the list of positive entries of ℓ sorted decreasingly, which here is ℓ with its trailing zeros removed. So $C(B(\lambda)) = R(C)$, and E is preserved by 1.1.

(ii) Row 1 of $R(C)$ has length $\lambda_1 - 1 > s$, so $(1, s) \in R(C)$ but $(0, s) \notin R(C)$: not an order ideal. For an arrangement $\ell = (\ell_0, \ell_1, \dots)$ of row lengths, the set of left-justified rows has energy $\sum_i (i \ell_i + \ell_i(\ell_i - 1)/2)$. The second sum does not depend on the order, and zero rows contribute

0. Let ℓ' be ℓ with entries 0, 1 swapped. Then $\sum_i i\ell'_i = \sum_i i\ell_i - (\lambda_1 - 1 - s) \leq \sum_i i\ell_i - 1$. By the rearrangement inequality, the decreasing arrangement $\ell^\downarrow$ of ℓ' satisfies $\sum_i i\ell_i^\downarrow \leq \sum_i i\ell'_i$. Now $C(B(\lambda))$ is the left-justified set with row lengths $\ell^\downarrow$ (zeros dropped at the end). Hence $E(B(\lambda)) \leq E(R(C)) - 1 = E(\lambda) - 1$. $\square$

1.4. Corollary. $E(B(\lambda)) \leq E(\lambda)$ always, and $E(B(\lambda)) = E(\lambda)$ iff $R(C(\lambda))$ is an order ideal, in which case $C(B(\lambda)) = R(C(\lambda))$. (Combine (i) and (ii); the two cases are exhaustive.)

2 Minimum energy and cyclicity (R2)

2.1. For finite $S \subset \mathbb{N}^2$ with $|S| = n$, let $N_{\geq d}(S) = \#\{x \in S : \text{diag}(x) \geq d\}$. Then $E(S) = \sum_{x \in S} \text{diag}(x) = \sum_{d \geq 1} N_{\geq d}(S)$, since a cell on diagonal e is counted once for each $d = 1, \dots, e$.

2.2. There are exactly T_d positions with diagonal $\leq d - 1$. So $N_{\geq d}(S) \geq n - T_d$, and also $N_{\geq d} \geq 0$. Put $\nu_d = \max(0, n - T_d)$ and $g_d(S) = N_{\geq d}(S) - \nu_d \geq 0$. Since $T_d \leq T_{k-1} < n$ for $d \leq k - 1$ and $T_d \geq T_k \geq n$ for $d \geq k$, we get $\nu_d = n - T_d$ for $1 \leq d \leq k - 1$ and $\nu_d = 0$ for $d \geq k$.

2.3. Set $E_{\min}(n) = \sum_{d \geq 1} \nu_d$ and $\Phi(S) = E(S) - E_{\min}(n) = \sum_{d \geq 1} g_d(S)$, a sum of non-negative integers.

2.4. Lemma. $\Phi(S) = 0$ iff $S = D_{k-2} \cup Q$ with Q a set of r positions on diagonal $k - 1$.

Proof. $\Phi = 0$ iff $g_d = 0$ for all d . For $d \geq k$ this says $N_{\geq k} = 0$: no cell has diagonal $\geq k$. For $1 \leq d \leq k - 1$ it says $\#\{x : \text{diag}(x) \leq d - 1\} = T_d$, i.e. all positions of diagonals $0, \dots, d - 1$ are in S . Taking $d = k - 1$, $D_{k-2} \subseteq S$. The remaining $n - T_{k-1} = r$ cells lie on diagonal $k - 1$. The converse is the same computation read backwards. $\square$

Lemma R2 (2.5). $\lambda \vdash n$ is cyclic iff $\Phi(\lambda) = 0$.

Proof. By Cell 1 (cited), the cyclic partitions are $\lambda(e) = (k - 1 + e_1, \dots, 1 + e_{k-1}, e_k)$ with $e \in \{0, 1\}^k$ and $\sum e = r$. Row i ($0 \leq i \leq k - 1$) of $\lambda(e)$ has length $k - 1 - i + e_{i+1}$. So $C(\lambda(e)) = D_{k-2} \cup \{(i, k - 1 - i) : e_{i+1} = 1\}$, which has $\Phi = 0$ by 2.4. Conversely, suppose $\Phi(\lambda) = 0$. By 2.4, $C(\lambda) = D_{k-2} \cup Q$. Set $e_{i+1} = [(i, k - 1 - i) \in Q]$. Then row i has length $k - 1 - i + e_{i+1}$ for $0 \leq i \leq k - 1$ and there are no further rows. So $\lambda = \lambda(e)$ with $\sum e = r$, and λ is cyclic by Cell 1. $\square$

2.6. Consequence. If $\Phi(\lambda) = 1$, then λ is not cyclic. Either $\Phi(B\lambda) = 1$ and $C(B\lambda) = R(C(\lambda))$, or $\Phi(B\lambda) = 0$ and $B\lambda$ is cyclic. (By 1.4 the energy stays the same or drops by ≥ 1 , and $\Phi \geq 0$.)

3 Partitions with $\Phi = 1$ (R3)

3.1. Let $\Phi(S) = 1$ for an order ideal S with $|S| = n$. Then there is exactly one d^* with $g_{d^*} = 1$, and $g_d = 0$ for $d \neq d^*$.

3.2. Case $d^* \geq k$. Here $g_d = 0$ for $d \leq k - 1$, so $D_{k-2} \subseteq S$ as in 2.4. For $d \geq k$, $N_{\geq d} = g_d$. $N_{\geq d}$ is non-increasing in d and $N_{\geq d^*} = 1$, so if $d^* > k$ then $N_{\geq k} \geq 1$, contradicting $g_k = 0$. Hence $d^* = k$, $N_{\geq k} = 1$ and $N_{\geq k+1} = 0$: exactly one cell lies off D_{k-1} , and it is on diagonal k . Diagonal $k - 1$ then holds $r - 1$ cells. Call this **case B**.

3.3. Case $d^* \leq k - 1$. Here $N_{\geq k} = 0$. Suppose first $d^* \leq k - 2$. Then $\#\{\text{diag} \leq d^* - 1\} = T_{d^*} - 1$ and $\#\{\text{diag} \leq d^*\} = T_{d^*+1}$ (because $g_{d^*+1} = 0$). So diagonal d^* would hold $d^* + 2$ cells, but it has only $d^* + 1$ positions. Hence $d^* = k - 1$. Diagonals $0, \dots, k - 3$ are full, and diagonal $k - 2$ has exactly one missing position $(i, k - 2 - i)$. The order-ideal property then excludes $(i, k - 1 - i)$ (whose left neighbour is $(i, k - 2 - i)$) and $(i + 1, k - 2 - i)$ (whose upper neighbour is $(i, k - 2 - i)$).

So diagonal $k - 1$ holds at most $k - 2$ cells. It holds $n - (T_{k-1} - 1) = r + 1$ cells, so $r \leq k - 3$. Call this **case A**.

Lemma R3 (3.4). *If $r \geq k - 2$, every $\lambda \vdash n$ with $\Phi(\lambda) = 1$ is of case B. In particular this holds for $n = T_k - 1$, where $r = k - 1$. (Immediate from 3.2 and 3.3.)*

3.5. Parametrising case B. Let $2 \leq r \leq k - 1$ and $h = k - r + 1$, so $2 \leq h \leq k - 1$. A case-B diagram is determined by the set $H \subseteq \{0, \dots, k - 1\}$ of rows of the h empty positions of diagonal $k - 1$ (“holes”) and the row $c \in \{0, \dots, k\}$ of the cell on diagonal k :

$$S(H, c) = D_{k-2} \cup \{(i, k - 1 - i) : 0 \leq i \leq k - 1, i \notin H\} \cup \{(c, k - c)\}.$$

Call (H, c) *admissible* if $(c \geq 1 \Rightarrow c - 1 \notin H)$ and $(c \leq k - 1 \Rightarrow c \notin H)$.

Claim (R3'): $S(H, c)$ is an order ideal iff (H, c) is admissible.

Proof. A cell of D_{k-2} has its up/left neighbours in $D_{k-3} \subseteq D_{k-2}$. A cell on diagonal $k - 1$ has its neighbours on diagonal $k - 2$, which is full. The cell $(c, k - c)$ has upper neighbour $(c - 1, k - c)$ (present iff $c \geq 1$ and $c - 1 \notin H$, required only if $c \geq 1$) and left neighbour $(c, k - 1 - c)$ (present iff $c \leq k - 1$ and $c \notin H$, required only if $k - c \geq 1$). $\square$

The partition with diagram $S(H, c)$ is written $\Lambda(H, c)$. Its row i ($0 \leq i \leq k$) has length $k - 1 - i + [i \notin H] + [i = c]$ (with row k of length $[c = k]$).

4 Exact dynamics in case B (R4)

4.1. By 1.1, R maps diagonal d to itself by row $i \mapsto (i + 1) \bmod (d + 1)$. So $R(S(H, c)) = S(H + 1, c')$, where $H + 1 = \{(a + 1) \bmod k : a \in H\}$ and $c' = (c + 1) \bmod (k + 1)$. (It fixes D_{k-2} setwise, permutes diagonal $k - 1$ with period k and diagonal k with period $k + 1$.)

4.2. Let $\lambda = \Lambda(H, c)$ with (H, c) admissible. By 2.6 and 1.4: if $(H + 1, c')$ is admissible, then $R(C(\lambda))$ is an order ideal (3.5), so $B\lambda = \Lambda(H + 1, c')$, again with $\Phi = 1$. Otherwise $R(C(\lambda))$ is not an order ideal, so the energy drops (1.4), so $B\lambda$ is cyclic (2.6). By induction on t :

$$d_B(\Lambda(H, c)) = \tau(H, c) := \min\{t \geq 1 : (H + t, (c + t) \bmod (k + 1)) \text{ not admissible}\}.$$

Here $H + t$ is taken mod k . For $t < \tau$ the partition $B^t \lambda = \Lambda(H + t, c + t)$ has $\Phi = 1$ and is not cyclic (2.5), and $B^\tau \lambda$ is cyclic. (Finiteness of τ follows from 4.4.)

4.3. Reduction to one hole. For $\alpha \in [0, k - 1]$ and $\gamma \in [0, k]$, the pair $(\{\alpha\}, \gamma)$ fails admissibility iff $\alpha = \gamma - 1$ or $\alpha = \gamma$. The side conditions $\gamma \geq 1$ and $\gamma \leq k - 1$ are automatic in these two equalities, because $0 \leq \alpha \leq k - 1$. So (H, γ) is admissible iff every $(\{a\}, \gamma)$, $a \in H$, is admissible. Hence $\tau(H, c) = \min_{a \in H} \tau_a$, where τ_a is the first $t \geq 1$ at which $w_t := \alpha_t - \gamma_t \in \{-1, 0\}$, with $\alpha_t = (a + t) \bmod k$ and $\gamma_t = (c + t) \bmod (k + 1)$.

4.4. Lemma. If $(\{a\}, c)$ is admissible, then $\tau_a = \min\{t \geq 1 : a + t \equiv 0 \pmod{k}, c + t \equiv 1 \pmod{k + 1}\} = 1 - c + (k + 1)m_a$, where $m_a = (c - 1 - a) \bmod k \in [0, k - 1]$.

Proof. Suppose $w_{t-1} \notin \{-1, 0\}$. We have $w_{t-1} \in [-k, k - 1]$. Four cases:

- (1) Neither α nor γ wraps ($\alpha_{t-1} < k - 1$, $\gamma_{t-1} < k$). Then $w_t = w_{t-1} \notin \{-1, 0\}$.
- (2) Only α wraps ($\alpha_{t-1} = k - 1$, $\gamma_{t-1} \leq k - 1$). Then $w_t = w_{t-1} - k$. Since $w_{t-1} \leq k - 1$, this is ≤ -1 , and it equals -1 iff $w_{t-1} = k - 1$, i.e. $\gamma_{t-1} = 0$, i.e. $(\alpha_t, \gamma_t) = (0, 1)$.
- (3) Only γ wraps ($\gamma_{t-1} = k$, $\alpha_{t-1} \leq k - 2$). Then $w_t = \alpha_{t-1} + 1 \geq 1$.

(4) Both wrap. Then $w_{t-1} = k - 1 - k = -1$, which is excluded.

So if t is the first failure time, w_{t-1} is admissible (for $t = 1$ by hypothesis), and only case (2) with $(\alpha_t, \gamma_t) = (0, 1)$ can fail. Conversely $(\alpha_t, \gamma_t) = (0, 1)$ gives $w_t = -1$, a failure. Hence τ_a is the least $t \geq 1$ with $a + t \equiv 0 \pmod{k}$ and $c + t \equiv 1 \pmod{k+1}$.

Solve by the Chinese remainder theorem ($\gcd(k, k+1) = 1$). Write $t = 1 - c + (k+1)m$. Since $k+1 \equiv 1 \pmod{k}$, the first congruence becomes $m \equiv c - 1 - a \pmod{k}$. So the solutions are $t = 1 - c + (k+1)(m_a + kq)$, $q \in \mathbb{Z}$. Put $t_0 = 1 - c + (k+1)m_a$. Then $t_0 \leq 1 + (k+1)(k-1) = k^2$, so $t_0 - k(k+1) \leq -k < 1$. And $t_0 \geq 1$: if $m_a \geq 1$ then $(k+1)m_a \geq k+1 > c$. If $m_a = 0$ then $a \equiv c - 1 \pmod{k}$; for $c \geq 1$ this means $a = c - 1$, which contradicts admissibility, so $c = 0$ and $t_0 = 1$. Hence $\tau_a = t_0$. $\square$

Lemma R4 (4.5). *For admissible (H, c) :*

$$d_B(\Lambda(H, c)) = 1 - c + (k+1) \min_{a \in H} ((c - 1 - a) \bmod k).$$

(From 4.2, 4.3 and 4.4.)

Remark. Exact check (not load-bearing): `code/check_c3.py` compares this formula with brute-force d_B for every case-B partition with $2 \leq r \leq k-1$, for $3 \leq k \leq K$ (output lines “(L)”). With $K = 10$ all match; see “How to verify” below.

5 Maximum over case B (R5)

5.1. The map $a \mapsto m_a = (c - 1 - a) \bmod k$ is a bijection of $\{0, \dots, k-1\}$, so the m_a ($a \in H$) are h distinct values. Admissibility says: $c \geq 1 \Rightarrow m_a \neq 0$ (since $a \neq c-1$), and $c \leq k-1 \Rightarrow m_a \neq k-1$ (since $a \neq c$ iff $c-1-a \not\equiv -1$).

5.2. The minimum of h distinct integers in an interval $[x, y]$ is at most $y - h + 1$. Hence:

- $c = 0$: $m_a \in [0, k-2]$, so $\min m_a \leq k-1-h$ and $d_B \leq 1 + (k+1)(k-1-h)$.
- $1 \leq c \leq k-1$: $m_a \in [1, k-2]$, so $d_B \leq 1 - c + (k+1)(k-1-h) \leq (k+1)(k-1-h)$.
- $c = k$: $m_a \in [1, k-1]$, so $\min m_a \leq k-h$ and $d_B \leq 1 - k + (k+1)(k-h) = (k+1)(k-1-h) + 2$.

Lemma R5 (5.3). *Over all case-B partitions of $n = T_{k-1} + r$ ($2 \leq r \leq k-1$), $\max d_B = (k+1)(r-2) + 2$. It is attained only by $c = k$ with $\{m_a\} = \{k-h, \dots, k-1\}$, i.e. $H = \{a = k-1-m\} = \{0, \dots, h-1\}$.*

Justification (as in the accepted artefact). This (H, c) is admissible: it needs $k-1 \notin H$, i.e. $h \leq k-1$, which holds. Here $h = k-r+1$ gives $(k+1)(k-1-h) + 2 = (k+1)(r-2) + 2$. The first two bullets of 5.2 are strictly smaller. $\square$

5.4. For $n = T_k - 1$ ($r = k-1$, $h = 2$, $k \geq 3$), the maximiser is $H = \{0, 1\}$, $c = k$. Its row lengths (3.5) are: row 0: $k-1$; row 1: $k-2$; row i for $2 \leq i \leq k-1$: $k-i$; row k : 1. That is,

$$\lambda_k^* = (k-1, k-2, k-2, k-3, \dots, 2, 1, 1),$$

with sum $(k-1) + (k-2) + T_{k-2} + 1 = T_k - 1$. Its value is $(k+1)(k-3) + 2 = k^2 - 2k - 1$. By 3.4 every $\Phi = 1$ partition of $T_k - 1$ is of case B. So λ_k^* is the unique partition of $T_k - 1$ with $\Phi = 1$ and $d_B = k^2 - 2k - 1$.

6 Lower bound for (b) (R6)

Lemma R6 (6.1). *For every $k \geq 3$, $d_B(\lambda_k^*) = k^2 - 2k - 1$ (5.4). So $D_B(T_k - 1) \geq k^2 - 2k - 1$.*

Direct check with 4.5. $m_0 = k - 1$ and $m_1 = k - 2$, so $d = 1 - k + (k + 1)(k - 2) = k^2 - 2k - 1$.
Examples: $k = 4$: $(3, 2, 2, 1, 1)$, $d = 7$. $k = 5$: $(4, 3, 3, 2, 1, 1)$, $d = 14$. $\square$

Open parts

The following are **OPEN**; no proof of them is part of this submission, and no conjectured answer is presented as established.

- (a) The upper bound $d_B(\lambda) \leq k^2 - 2k - 1$ for every non-triangular n of rank $k \geq 4$ (every $T_{k-1} < n < T_k$ and every $\lambda \vdash n$): OPEN.
- (b) The upper half of (b), $D_B(T_k - 1) \leq k^2 - 2k - 1$; so the equality $D_B(T_k - 1) = k^2 - 2k - 1$ is NOT established: OPEN. Only the lower bound $D_B(T_k - 1) \geq k^2 - 2k - 1$, $k \geq 3$ (Lemma R6), is proved.
- (c) The set of partitions of $T_k - 1$ attaining $D_B(T_k - 1)$: OPEN.

Numerical evidence (not proof). The shipped exact checker `code/check_c3.py`, run as `python3 check_c3.py 10`, enumerates every partition of every n with $T_{k-1} < n < T_k$, $2 \leq k \leq 10$, and computes d_B by exact cycle detection. Its output (reproduced below from the run recorded in `verify/logs/check_c3.K10.txt`) reports $D_B(n) \leq k^2 - 2k - 1$ for all such n with $4 \leq k \leq 10$, and $D_B(T_k - 1) = k^2 - 2k - 1$ for $4 \leq k \leq 10$. This concerns exactly those finitely many n ; it proves nothing for $k \geq 11$ and is not part of the proof of any claim above.

k	2	3	4	5	6	7	8	9	10
$T_k - 1$	2	5	9	14	20	27	35	44	54
$D_B(T_k - 1)$ (computed)	0	3	7	14	23	34	47	62	79
$k^2 - 2k - 1$	-1	2	7	14	23	34	47	62	79

For $k = 2, 3$ the computed values differ from $k^2 - 2k - 1$. The checker's final line was ALL OK.

How to verify

- Command (stdlib-only Python 3, exact integer arithmetic), from the directory `submission/code/`:
`/usr/bin/time -p python3 check_c3.py 10`
- Expected output: last line ALL OK.
- Measured runtime: `real 26.93 s (user 26.66, sys 0.14)`, Python 3.14.0, output ALL OK; well under the 10-minute limit.
- The check is not load-bearing for Lemmas R1–R6; it checks Lemma R4's formula (lines "(L)") and $d_B(\lambda_k^*) = k^2 - 2k - 1$ for $4 \leq k \leq 10$ only.

Limitations

- Only the lower half of (b) is established. (a), the upper half of (b) and (c) are OPEN.
- Lemma R2, and hence everything after it, relies on the Cell 1 classification of cyclic partitions, cited as an earlier cell result and not re-proved here.
- The accepted artefact also contains further material (its sections 7–9: an explicit family of partitions with $d_B = k^2 - 2k - 1$, partial upper-bound analysis, and maximiser computations). That material was not gated and is deliberately not included here as a result.
- The computation above is numerical evidence over a finite range only.